

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

8876/01

Paper 1 Multiple Choice

13 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **15** printed pages and **1** blank page.



1 Which part of the cell is often continuous with the rough endoplasmic reticulum?

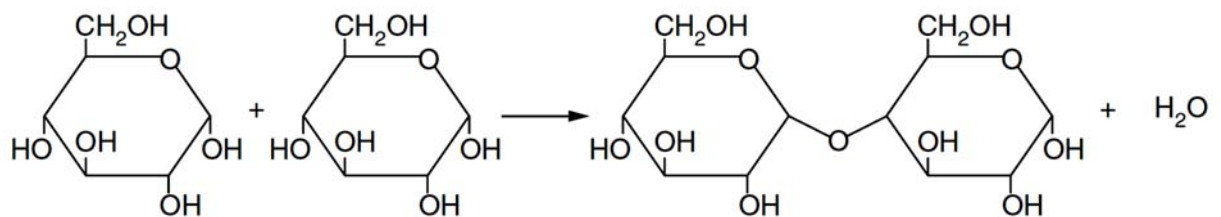
- A cell surface membrane
- B Golgi apparatus
- C mitochondrion
- D** nuclear envelope

2 Which structures are found in eukaryotic cells?

- 1 70S ribosomes
- 2 80S ribosomes
- 3 circular DNA
- 4 linear DNA

- A 1 and 2
- B 1 and 3
- C 2 and 4
- D** 1, 2, 3, and 4

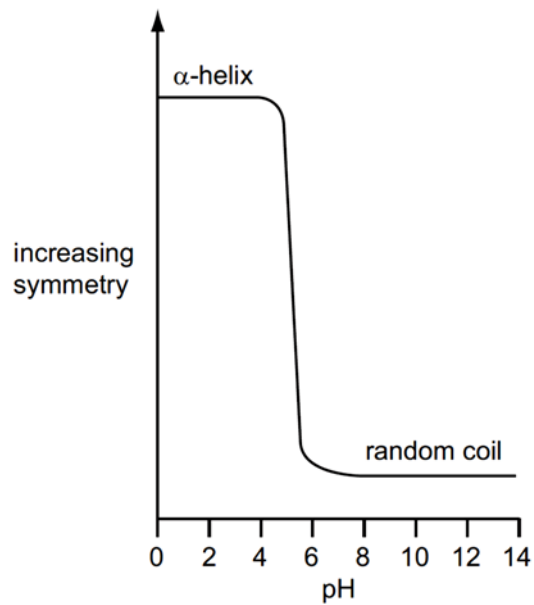
3 The diagram shows a reaction resulting in the formation of a bond between two molecules.



Which bond is formed and what is the type of reaction?

	bond formed	type of reaction
A	glycosidic	condensation
B	glycosidic	hydrolysis
C	peptide	condensation
D	peptide	hydrolysis

- 4 The graph shows the effect of pH on the structure of a protein which consists entirely of repeating residues of one amino acid.

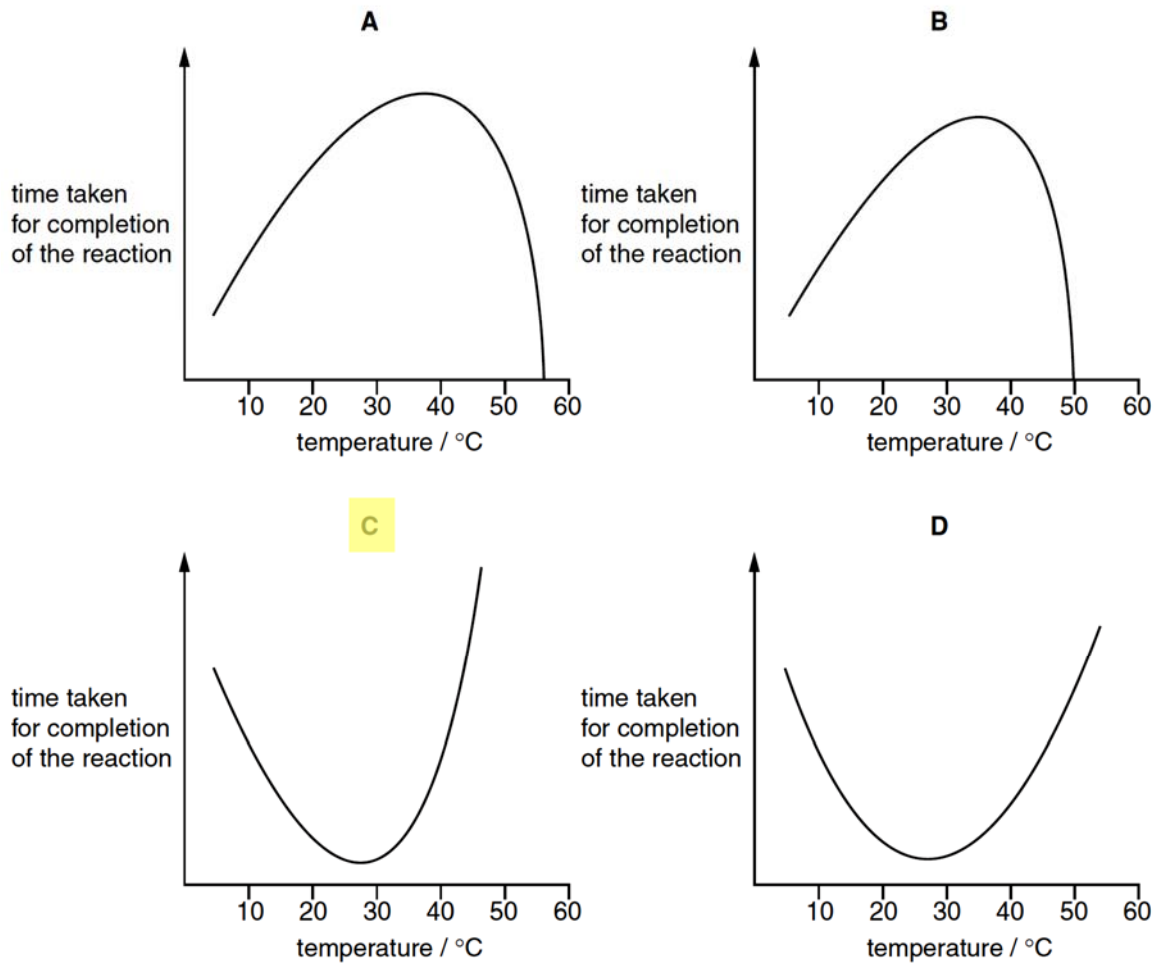


Which statement is true?

- A At pH2 the protein has lost its secondary structure.
- B At pH2 the protein has lost its tertiary structure.
- C At pH10 the protein has lost its primary structure.
- D** At pH10 the protein has lost its secondary structure.

- 5 An enzyme is completely denatured at 50 °C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?



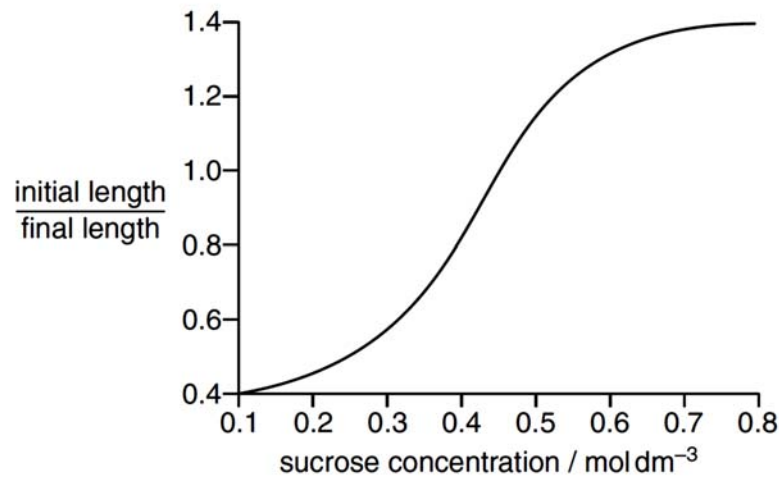
- 6 Which bonds are the last to break when an enzyme is heated?

- A** disulfide
- B** hydrogen
- C** hydrophobic interactions
- D** ionic

- 7 Increasing which type of bond helps to increase the fluidity of the cell surface membrane?

- A** C — O — C
- B** C — C
- C** C = C
- D** hydrogen

- 8 Strips of plant tissue were immersed in a range of sucrose solutions of different concentrations. Their lengths were measured before immersion and after 30 minutes in the different solutions. The graph shows the ratio of initial length to final length.



Which concentration of sucrose solution, in mol dm⁻³, has the same water potential as the cell sap before immersion?

- A 0.1
 - B 0.25
 - C 0.45**
 - D 0.8
- 9 Which of the following best illustrates totipotency of stem cells?
- A A somatic cell isolated from a root tip develops into a normal adult plant.**
 - B Stem cells are able to divide indefinitely.
 - C Mesenchymal stem cells can differentiate into an extensive range of cell types, including bone cells, cartilage cells, muscle cells and fat cells.
 - D The replacement of the nucleus of an unfertilised egg with that of a pancreatic cell converts the egg into a pancreatic cell.
- 10 RNA is involved in the process of protein synthesis.

Which of the following descriptions is true about RNA in eukaryotes?

- A rRNA, which is coded by genes found in nucleolus, associates with ribosomal proteins in the cytoplasm to form ribosomal subunits.
- B Functional mRNA is formed as a result of post-transcriptional modifications of primary RNA transcript in the nucleus.**
- C The ribonucleotide sequence of tRNA molecules allows extensive folding and inter-strand complementarity to generate a three-dimensional structure.
- D RNA is synthesised in the nucleus only.

11 The mechanism of action of four drugs that inhibit DNA replication is stated below:

- Drug 1 inhibits the action of DNA ligase.
- Drug 2 resembles the shape of a DNA nucleotide.
- Drug 3 attaches irreversibly to the DNA molecule.
- Drug 4 binds irreversibly to the active site of DNA polymerase.

Which option correctly matches the drug(s) to the effect on DNA replication?

	Daughter strands of varying lengths are synthesized.	Only fragments are synthesized at the end of replication process.	Phosphodiester bonds cannot be formed.	Template strand becomes inaccessible by the enzyme.
A	4	3	2	1
B	2	4	3	1
C	2	1	4	3
D	3	2	1	4

12 Which of the following statement(s) is/are **not** true of the translation process in all eukaryotes?

- 1 Polypeptides are only synthesised in the cytosol.
- 2 Amino acids are linked by the formation of peptide bonds catalysed by a ribozyme.
- 3 Ribosomes contain an amino-acyl tRNA site that is occupied by the initiator tRNA attached to methionine.
- 4 Amino-acyl tRNA synthetase attaches an amino acid to the 5' end of a tRNA molecule.

- A** 1, 3 and 4 only
B 2, 3 and 4 only
C 2 and 4 only
D 1 only

- 13 In 1985, it was discovered that a bacterium, *Mycoplasma capricolum*, used a deviant genetic code. The codon UGA resulted in the addition of tryptophan to the growing polypeptide chain.

A short sequence of nucleotides was synthesised with the following base sequence:

3' CTGGCAACTATTTCAACTCATATC 5'

How many peptide bonds would be formed by ribosomes when this sequence undergoes transcription and translation in *Mycoplasma capricolum* and a human liver cell?

	<i>Mycoplasma capricolum</i>	Human liver cell
A	1	0
B	2	1
C	3	2
D	4	3

- 14 The RNA triplet UAG acts as a stop codon terminating the synthesis of a polypeptide. The diagram shows a strand of DNA which codes for four amino acids.

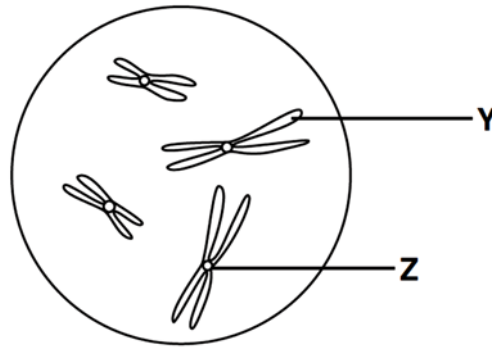
Where would a mutation, introducing a thymine nucleotide, result in the termination of translation?

T C C A C T C G A T G C
 ↑ ↑ ↑ ↑
 A B C D

- 15 Which of the following describes aneuploidy?

- A nonsense mutation
- B single base substitution
- C translocation
- D trisomy 21 i.e. Down's syndrome

- 16 The diagram shows chromosomes in a nucleus.



What are Y and Z?

	Y	Z
A	centromere	centriole
B	centromere	chromatid
C	chromatid	centriole
D	chromatid	centromere

- 17 Gene mutations in either the *BRCA1* or the *BRCA2* genes are responsible for the majority of hereditary breast cancer in humans.

The proteins produced by the two genes migrate to the nucleus where they interact with other proteins, such as those produced by the tumour suppressor gene, *p53*, and the DNA repair gene, *RAD51*.

Which combination of gene activity is most likely to result in breast cancer?

key:

✓ = gene produces normal protein ✗ = gene produces abnormal protein or no protein

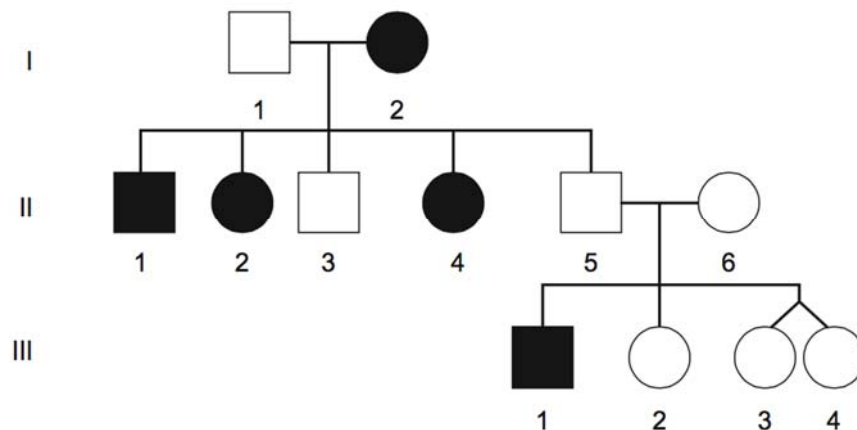
	gene		
	<i>BRCA1 or BRCA2</i>	<i>p53</i>	<i>RAD51</i>
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✗

- 18 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

- A one gene with a pair of co-dominant alleles
B one gene with multiple alleles
 C sex linkage of the allele for grey coat colour
 D two genes, each with a dominant and recessive allele
- 19 The pedigree below showed that the mode of inheritance of this disease is as supported by



- A autosomal dominant, individuals I-2 with offspring II-1, II-2 and II-4
 B autosomal recessive, individuals I-1, I-2 with offspring II-3 and II-5
C autosomal recessive, individuals II-5, II-6 with offspring III-1
 D sex-linked dominant, offspring II-2 and II-4

- 20** Fruit flies (*Drosophila*), homozygous for long wings, were crossed with fruit flies homozygous for vestigial wings. The F_1 and F_2 generations were raised at three different temperatures.

At each temperature, the F_1 generation all had long wings.

The table shows the results in the F_2 generation.

temperature / °C	result
21	$\frac{3}{4}$ long wings, $\frac{1}{4}$ vestigial wings
26	$\frac{3}{4}$ long wings, $\frac{1}{4}$ intermediate wing length
31	all long wings

Which statement explains these results?

- A** Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
- B** Long wing and vestigial wing illustrate codominance at 26°C.
- C** Long wing is dominant at higher temperatures but vestigial wing is dominant at lower temperatures.
- D** Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.
- 21** In respiration, the enzyme hexokinase uses ATP to transfer a phosphate group to glucose to form glucose-6-phosphate.

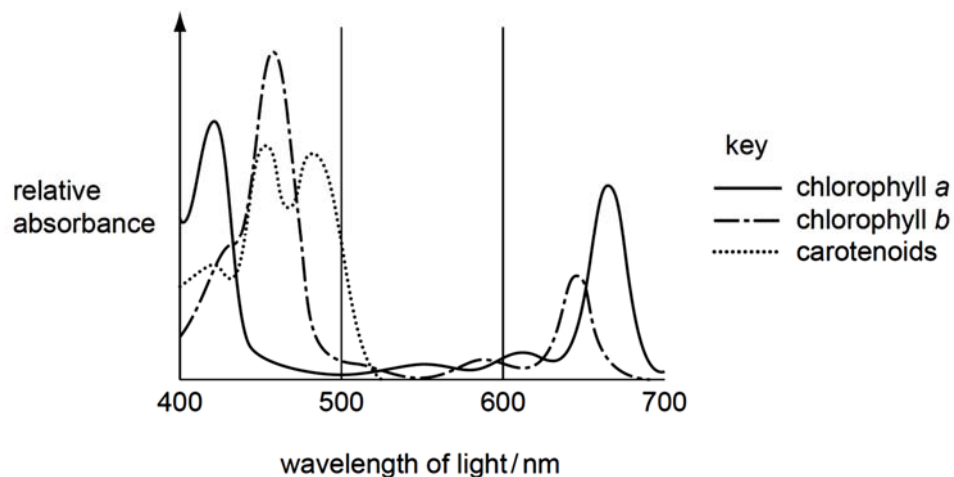
If a cell only has glucose available for energy and the activity of hexokinase is suddenly inhibited in this cell, which of the following will occur?

- 1 The cell will not be able to produce pyruvate through glycolysis.
 - 2 Respiratory processes in the mitochondria would not proceed.
 - 3 The use of oxygen by the cell will decrease.
- A** 1, 2 and 3
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 and 3 only

22 Which of the following statements explains the mechanism for chemiosmosis in the synthesis of ATP?

- A The energy released by the reduction and subsequent oxidation of components of the electron transport chain is transferred to ATP synthase for the synthesis of ATP.
- B Phosphorylation of ADP is linked to the proton gradient established by the electron transport chain.**
- C The difference in pH between the intermembrane space and the cytosol drives the formation of ATP.
- D The flow of H^+ through ATP synthase into the intermembrane space drives the synthesis of ATP.

23 The graph shows the absorption spectra of some pigments found in chloroplasts.

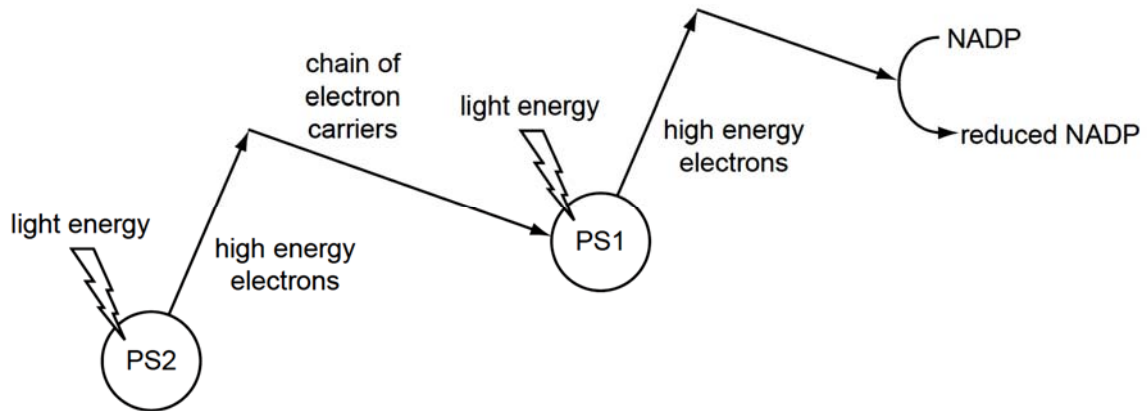


Which statements are correct?

- 1 Having several pigments, rather than one, increases the efficiency of photosynthesis.
- 2 Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
- 3 Photosynthesis will be fastest in light at the red end of the spectrum, as red light has higher energy than blue light.
- 4 Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.

- A 1 and 2 only
- B 1 and 3 only
- C 1, 2 and 4 only**
- D 2 and 4 only

- 24 The diagram shows some of the processes in the light-dependent stage of photosynthesis.



For the light-dependent stage to continue, photosystem two (PS2) must gain electrons. Where do these electrons come from?

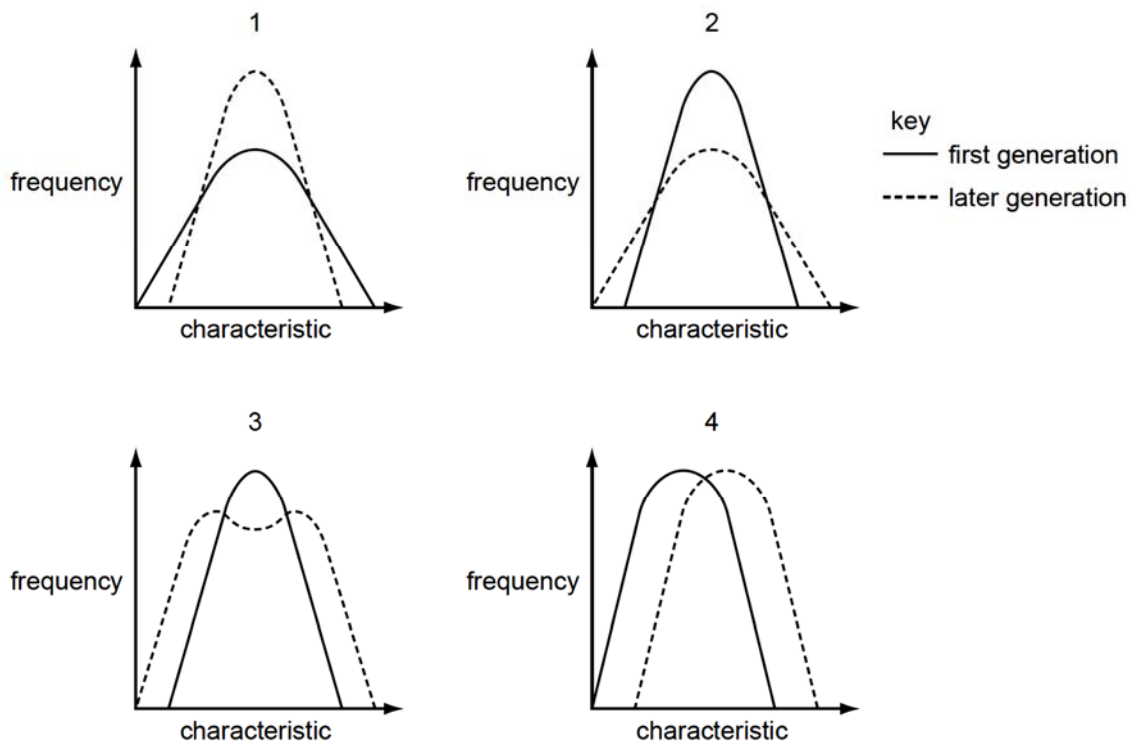
- A electron carriers
 - B reduced NADP
 - C photolysis of water**
 - D the formation of ATP
- 25 Which statements are true about Darwinian evolutionary theory?
- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
 - 2 In competition for survival, the more aggressive animals are more likely to survive.
 - 3 Species perfectly adapted to a stable environment will continue to evolve.
 - 4 Variation between individuals of a species is essential for evolutionary change.
- A 1, 2 and 4 only
 - B 2 and 3 only
 - C 3 and 4 only
 - D 4 only**

- 26** The Galapagos Islands are a group of volcanic islands in the eastern Pacific Ocean, about 1000 km from South America. Thirteen species of finch are found on the islands; they resemble each other closely but differ in their feeding habits and in shape of their beaks.

Assuming that an ancestral stock of finches came from the mainland, what is the most likely explanation for the existence of similar but distinct species of Galapagos finches?

- A** Finches developed different kinds of beaks in order to feed on different kinds of food.
- B** Finches evolved separately according to the habitat in which they settled.
- C** Finches from the mainland bred with population of a related species on the island to produce new genotypes.
- D** Finches in similar habitats face similar selection pressures thus evolve to become very similar species.

- 27** The graphs show frequency against a measured characteristic in the first and later generation of an organism.



Which graph represents each type of natural selection?

	directional	disruptive	stabilising
A	1	2	3
B	2	3	4
C	3	1	2
D	4	3	1

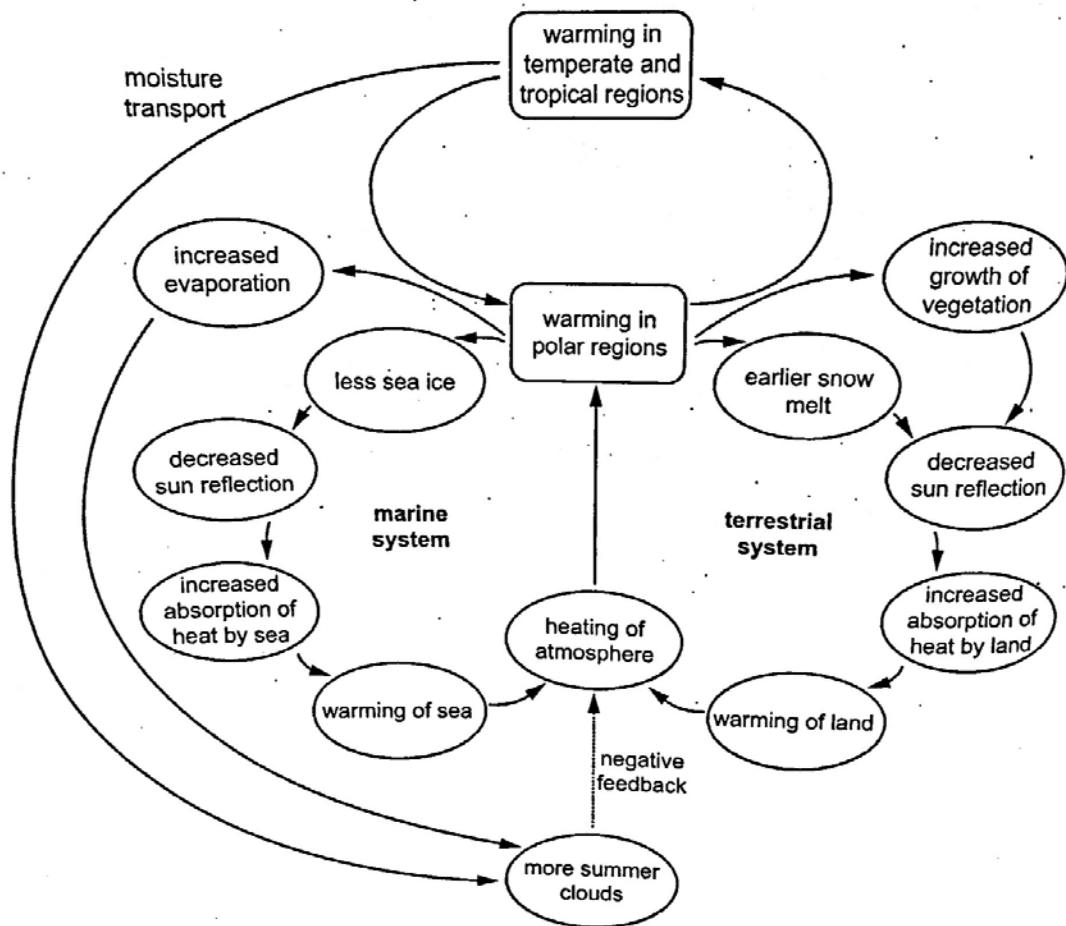
28 What is the impact of global warming on plants?

- A In colder regions, a warmer climate may allow people to grow new crops.
- B Global warming beyond optimal growth temperatures encourages plant growth.
- C Temperate plants will shift to tropical regions.
- D Production of all crops are higher due to higher temperatures.

29 Which of the following correctly shows the effects of climate change on coral reefs and associated ecosystems?

	Average number of zooxanthellae in each polyp	Mass of basal plate of hard corals	Diversity of catch from nearby fisheries
A	Decreased	Decreased	Decreased
B	Decreased	Unaffected	Increased
C	Increased	Decreased	Decreased
D	Increased	Unaffected	Increased

- 30** The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

- A** Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.
- B** Increased evaporation leads to more rainfall, which absorbs heat from the land and the sea.
- C** Melting sea ice causes more cloud formation, which increases absorption of heat in the atmosphere.
- D** Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.

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